

Xiangyu Ma

✉ MXY1963937485@hotmail.com |  [Personal Website](#) |  [GitHub](#)
Changchun, Jilin, China

RESEARCH INTERESTS

Quantum information theory; Bell nonlocality and multipartite entanglement; quantum algorithms; quantum many-body physics; open and nonequilibrium quantum systems.

EDUCATION

- **Jilin University** Sept. 2021 – Jun. 2026
B.Eng. in Transportation Engineering Changchun, China
 - GPA: 84.16/100
 - **Physics and Mathematics:** Quantum Mechanics (95.1), Thermodynamics and Statistical Physics (80.2), Calculus (93), Mathematical Modeling (95), University Physics (85.6), Electrodynamics(coming soon), Lie Groups and Lie Algebras(coming soon), and Methods of Mathematical Physics II (86.2).
 - **Computing and Data Science:** University Computer (91), Theory and Practice of Artificial Intelligence (96.9), Theory and Method of Transportation Big Data (86), Course Experiment of Transport Big Data Theory and Method (90).
- **Jilin University** Sept. 2025 – Jun. 2026
Minor in Quantum Information Science Changchun, China
 - Selected Coursework: Introduction to Quantum Information, Laser Spectroscopy, Quantum Optics, Quantum Computing, and Quantum Precision Measurement.

RESEARCH EXPERIENCE

- **Quantum Nonlocality and Multipartite Entanglement in the $(3, 2, 2)$ Scenario** July 2025 – Present
University of Tokyo
Supervisor: Prof. Naomichi Hatano; Research Mentor: Asst. Prof. Jaeha Lee
 - Developed a systematic framework for tripartite Bell-violation mechanisms in the $(3, 2, 2)$ scenario, showing how entanglement structure and local measurement geometry jointly determine Mermin and Svetlichny violations.
 - Classified correlations across fully separable, biseparable, and genuinely multipartite-entangled sectors; derived a CHSH-type factorization for biseparable states, proved the Svetlichny bound $|S| \leq 4$, and characterized generalized GHZ, W, and representative mixed-state families.
 - Analyzed the block structure and singular-value spectra of multipartite correlation tensors to identify measurement-sensitive sectors and explain violation enhancement and branch switching in GHZ–W mixtures.
 - Developed reproducible Python workflows combining Monte Carlo sampling, analytical and numerical parameter optimization, and semidefinite programming based on the NPA hierarchy to map violation regions and verify analytical boundaries.
- **Nuclear Matter Phase Structure Study** June 2023 – Aug. 2023
Institute of Modern Physics, Chinese Academy of Sciences
Supervisor: Hao Qiu
 - Studied the phase structure of nuclear matter, with emphasis on quark confinement, quark–gluon plasma, and critical behavior.
 - Conducted data analysis and numerical model implementation using CERN ROOT, including exploratory work on hypernucleus reconstruction.

SKILLS

- **Programming and Scientific Computing:** Python (NumPy, SciPy, CVXPY, Matplotlib), C/C++, MATLAB, CERN ROOT
- **Numerical Methods:** Monte Carlo sampling, analytical and numerical parameter optimization, semidefinite programming, and the NPA hierarchy
- **Machine Learning and Data Analysis:** Feedforward neural networks, GRU and LSTM models, clustering, time-series forecasting, and model evaluation
- **Research Tools:** $\text{\LaTeX}$, Git, and COMSOL
- **Languages:** Chinese (native), English (TOEFL 89)

HONORS AND AWARDS

- **Second-Class Scholarship (Rank 7/74)** Apr. 2026
Jilin University
- **College Excellent Student (Rank 7/74)** Apr. 2026
Jilin University
- **Provincial Second Prize, National College Students Mathematical Contest in Modeling** Dec. 2023
Chinese Society of Industrial and Applied Mathematics
- **Second Prize, China Undergraduate Physics Tournament (CUPT)** Dec. 2022
College of Physics, Jilin University

SELECTED ACADEMIC PROGRAMS

- **Physics of Open Systems: Resonance, Symmetry and Topology**
University of Tokyo
 - Topics: Open quantum systems, non-Hermitian physics, and Floquet topology.

Aug. 2025


- **International Summer School for Graduate Students on Optics**
Fudan University
 - Topics: Quantum optics, nanophotonics, and nonlinear and ultrafast optics.

Aug. 2024